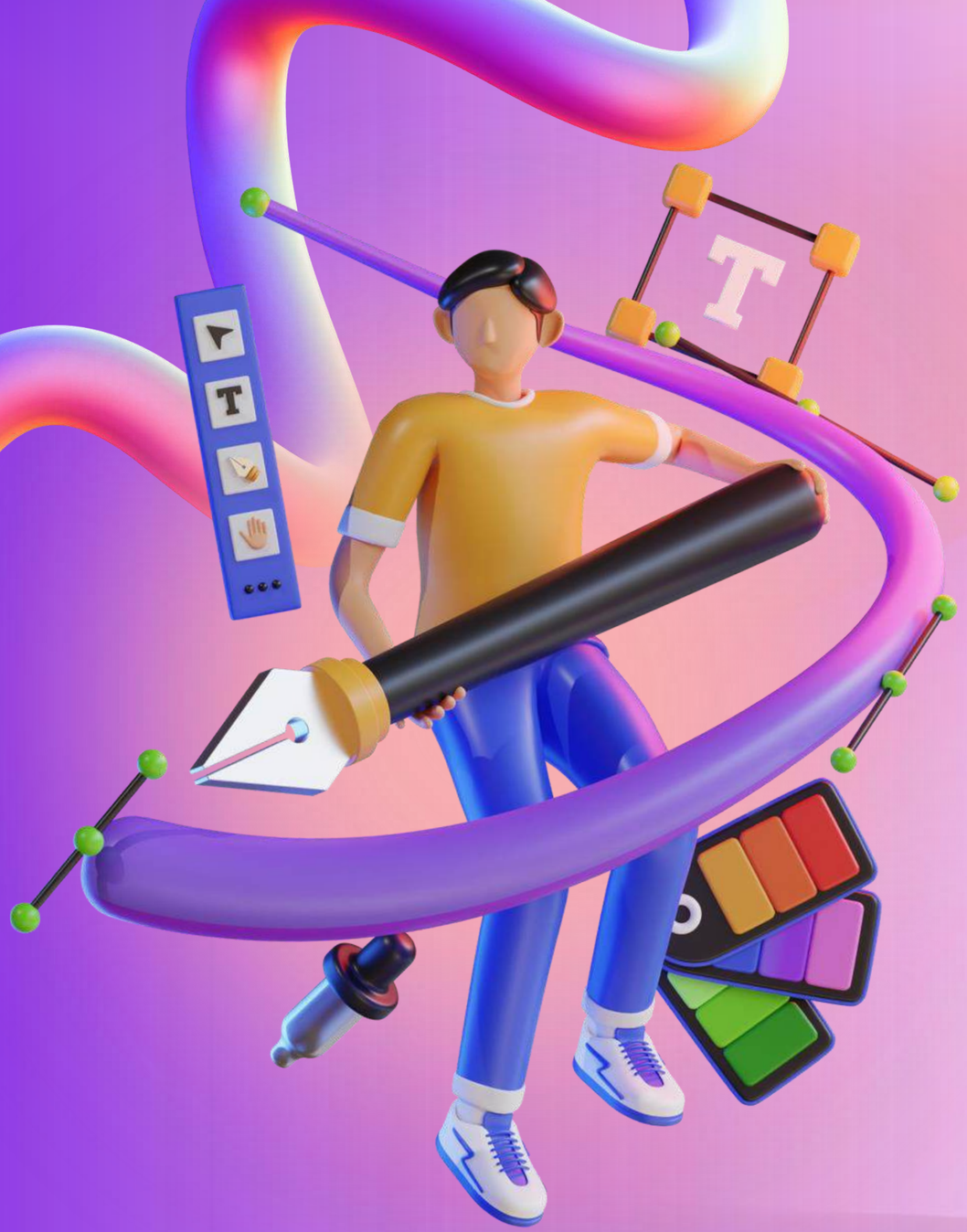


IYSA Indonesian Young Scientist Association

NICYMS National Invention Competition for Young Moslem Scientists

UIN Sunan Gunung Djati Bandung

Science In Islam

Integration of Islamic Waqf & Charity Big Data for Sustainable Sosio-Ecological Planning

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Our Team



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Introduction

✦ Charity in Islam

- Rooted from the Arabic word ʿamala (عَمِلَ) – meaning "to act or do".
- Encompasses waqf, zakat, and alms — spiritual and strategic tools.
- Historically used for education, health, infrastructure, and economy.

✦ Modern Challenges

- Disconnected, scattered charity data across institutions.
- Lack of integrated mapping, analytics, or planning tools.
- Missed potential to support data-driven social justice and development.

✦ Strategic Opportunity

- Big Data + GIS = Real-time insights, spatial precision, policy support.
- Revitalizing productive waqf for today’s socio-ecological needs.
- Aligns with SDGs: poverty, education, sustainable cities.

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Research Objectives



1. Developing a Big Data and GIS-based information system design that is capable of collecting, mapping, and analyzing waqf, zakat, and alms data spatially and temporally.

2. Applying the system as a decision-making tool for socio-ecological planning relevant to SDGs targets 1, 4, and 11.

3. Developing a framework for revitalizing technology-based productive waqf as an integrative model between Islamic values and data-based sustainable development principles.

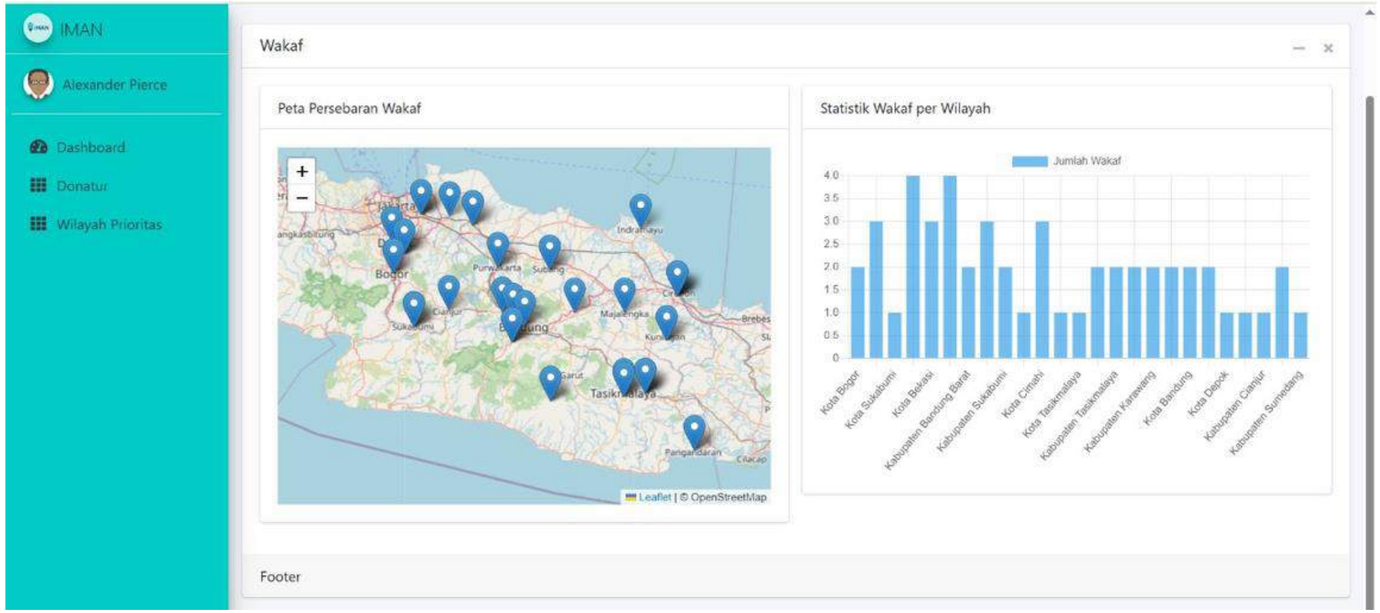
Method



Literature Review & Needs Analysis → Data Collection and Integration → System Design & Implementation → System Testing &Validation

This study adopts a Research and Development (R&D) approach using an iterative prototyping model to design and implement an integrated information system for Islamic philanthropy based on Big Data technology and a simplified Geographic Information System (GIS). This system is developed as a web-based application that is able to integrate and visualize waqf, zakat, and alms data spatially and temporally.

Result



Spatial mapping, with Leaflet showing the successful visualization of more than 20 locations of waqf, zakat, and alms spread across West Java. This facilitates the analysis of distribution and identification of areas that require priority allocation.

Analytical dashboard visualization, providing real-time information on asset value, distribution intensity per region, and contribution trends, thus supporting the evaluation process of service gaps.

Result



requiring in-depth technical expertise.

Laravel's modular structure and use of RESTful APIs, ensure system scalability, allowing for advanced integrations such as mobile app support, government data connections, or synchronization with external mosque-based platforms.

Data management, performed through a relational database with consistent foreign key relationships, enables reliable tracking of distribution history and data analysis.

With the use of real data and role-based authentication features, the system realistically represents the workflow of Islamic philanthropy. Evidence-based visualization facilitates the identification of distribution inequalities and supports fairer and more strategic resource allocation planning.



Conclusion

This experiment shows that the integration of GIS technology and Big Data has great potential in optimizing Islamic philanthropy management. Through the digitization of waqf, zakat, and sedekah data, policy makers gain important insights into spatial inequality and contribution trends, which ultimately strengthen data-driven governance and social justice.

The technologies used, namely Laravel, Leaflet, and Chart.js, have proven to be reliable as the foundation of a socio-ecological intelligence system based on Islamic values. Future developments can be directed at direct integration with official APIs such as BAZNAS, providing a mobile interface for nazhir in remote areas, and implementing predictive analytics based on artificial intelligence to support more adaptive social planning



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Thank You For Attention